Degree in Chemical Engineering

Zeolite Framework Analysis: Advancing Separation Processes with Machine Learning

Internship Report in Academic Environment

of

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Abstract

In the pursuit of more efficient and environmentally sustainable separation processes in the chemical industry, this project leverages Scientific Machine Learning techniques to predict the framework density of zeolites based on their composite building units. Zeolites, with their microporous crystalline structures, are critical in adsorption-based separation processes. However, traditional methods of predicting their framework density are time-consuming and resource-intensive. This project introduces an innovative approach that integrates machine learning with chemical engineering principles, aiming to understand the zeolite framework to predict material properties for adsorption.

The core methodology involved the creation of a comprehensive database of zeolite properties, obtained through web scraping from the International Zeolite Association website. A unique fingerprint was generated for each zeolite, capturing the structural characteristics of their composite building units. These fingerprints were used as input to train a neural network model to predict framework density, offering a rapid and efficient alternative to experimental methods. The model was trained using a 5-fold cross-validation process, and while the results demonstrated the potential of this approach, some discrepancies were observed. Outliers in the predictions were primarily attributed to the limited representation of certain framework densities in the training data and the structural complexity of some zeolites.

The findings suggest that incorporating natural tiling fingerprints could enhance the model's accuracy, especially for zeolites not fully captured by composite building units based representations. This project provides a valuable contribution to the development of more efficient zeolite-based separation processes, with implications for both industrial applications and environmental sustainability.

Keywords: Zeolites, Framework Density, Scientific Machine Learning, Composite Building Units

Resumo

Na procura de processos de separação mais eficientes e sustentáveis na indústria química, este projeto recorre a técnicas de *Scientific Machine Learning* com o intuito de prever a densidade estrutural de zeólitos, com base nos seus *composite building units*. Zeólitos, com as suas estruturas cristalinas microporosas, são essenciais em processos de separação baseados em adsorção. No entanto, métodos tradicionais para prever as suas densidades estruturais são demorados e consomem variados recursos. Este projeto introduz uma nova abordagem que integra *Machine Learning* com princípios de engenharia química, visando entender a estrutura de zeólitos para prever propriedades materiais para adsorção.

A metodologia do projeto envolveu a criação de uma base de dados das propriedades de zeólitos, obtidas a partir de *Web Scraping* do site da *International Zeolite Association*. Uma *fingerprint* única foi criada para cada zeólito, capturando as características estruturais dos seus *composite building units*. Estas *fingerprints* foram usadas como *input* para treinar um modelo de rede neuronal, para prever a framework density de zeólitos, oferecendo uma alternativa rápida e eficiente a métodos experimentais. O modelo foi treinado usando o método de *5-fold cross-validation*, e, enquanto que os resultados demonstraram o potencial desta abordagem, algumas discrepâncias foram observadas. *Outliers* nas previsões foram atribuídos principalmente à representação limitada de certas densidades estruturais nos dados de treino do modelo e à complexidade estrutural de certos zeólitos.

Os resultados sugerem que a incorporação de fingerprints baseados em natural tiling, outra forma de exprimir a estrutura de zeólitos, poderia melhorar a precisão do modelo, especialmente para zeólitos que não são completamente caracterizados por representações baseadas em composite building units. Este projeto oferece uma contribuição valiosa para o desenvolvimento de processos de separação com zeólitos mais eficientes, tendo implicações tanto para aplicações industriais quanto para a sustentabilidade ambiental.

Palavras-chave: Zeólitos, Densidade Estrutural, *Scientific Machine Learning*, *Composite Building Units*

Declaration

Mariana Faria Pereira, hereby declares, under word of honor, that this work is original and that all non-original contributions are indicated, and due reference is given to the author and source.

Mariana Pereira

(Mariana Pereira)

Porto, 9th of September 2024

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Notation and Glossary

tanh	hyperbolic tangent

List of Acronyms

ANN	Artificial Neural Network
CBU	Composite Building Unit
CNN	Convolutional Neural Network
FD	Framework Density
IZA	International Zeolite Association
ML	Machine Learning
MSE	Mean Squared Error
SciML	Scientific Machine Learning
SGD	Sustainable Development Goal

1 Introduction

1.1 Framework and Presentation of the Project

The problem under study focuses on the recognition of structures of adsorbents to predict properties faster and more efficiently. Zeolites, with their microporous crystalline structures, offer significant potential for such application due to their ability to selectively adsorb specific molecules based on their framework density (FD). However, predicting the FD of zeolites, which is crucial for optimizing their performance in separation processes, remains a challenging task. Traditional methods for predicting FD are time-consuming and resource-intensive, relying heavily on trial and error during the synthesis of new zeolites.

To overcome these challenges, this project introduces an innovative approach that leverages Scientific Machine Learning (SciML) techniques, particularly the use of molecular fingerprinting. In molecular systems, techniques such as Morgan fingerprints have revolutionized the prediction of molecular properties by capturing essential structural features of molecules. These fingerprints are commonly used as input for SciML models, leading to accurate and efficient property predictions. However, despite the widespread success of fingerprinting in molecular systems, its application to zeolites remains unexplored. This project aims to fill that gap by adapting molecular fingerprinting techniques to create comprehensive "fingerprints" for zeolites, based on their composite building units (CBUs). The integration of these CBU fingerprints with SciML represents a novel method for the rapid screening and optimization of zeolite properties, offering a more efficient alternative to traditional experimental methods. To support this innovation, a large-scale database of zeolite properties was constructed through web scraping techniques from the International Zeolite Association (IZA) website.

1.2 Contribution of the author for the work

My contributions to this dissertation were the research of the theme, the development and writing of the Python codes for the web scraping, the creation of the CBU fingerprints and the SciML techniques. Some native functions and libraries of Python were used to facilitate the development, all of those are referred to in the text. On the other hand, I constructed, analyzed the codes, strategies and tools proposed here.

1.3 Outline

This work is organized into six main parts:

The first chapter, Introduction, presents the project, outlining the motivation, objectives, and its importance. It also provides an overview of the methods used.

In the following chapter, Context and state of the art, presents a literature review, summarizing relevant research on zeolite characterization and prediction methods.

In chapter three, Methodology, the methodology implemented throughout the development of this work is explained. This chapter is divided into two main sections: Database and Artificial Neural Network, the last one being organized into three subsections that detail the construction, training, and evaluation of an Artificial Neural Network designed to predict a single-valued output.

The fourth chapter discusses the results obtained, including the performance of the predictive model and its limitations.

The Conclusions chapter summarizes the key findings and the practical contributions of the project. It also provides proposals for future research.

Finally, the Evaluation of the performed work presents the objectives achieved, personal learning outcomes, and its alignment with Sustainable Development Goals (SDGs) for sustainable industrial practices.

2 Context and state of the art

Separation processes play a crucial role in our society, accounting for 10-15% of global energy consumption (Sholl & Lively, 2016) and 40-90% of industrial capital and operating costs (de Haan et al., 2020). According to Pérez-Botella et al. (2022), many existing chemical processes fail to produce products with the desired purity for immediate use. This is particularly true for petrochemical products, which require post-production purification steps (de Haan et al., 2020). Beyond chemically produced substances, the separation and purification of naturally occurring mixtures, such as air, is of significant environmental, economic, and/or practical importance (Erans et al., 2022; Ackley, 2019). These processes have historically focused primarily on efficiency and cost-effectiveness, often overlooking environmental impacts. However, with the growing concerns of climate change and resource scarcity, it is imperative to prioritize environmentally friendly separation technologies (Sholl & Lively, 2016).

Adsorption is an alternative separation technology used for various mixtures. It exploits the ability of certain solids to preferentially concentrate specific substances from gaseous or liquid solutions onto their surfaces. Over the past few decades, adsorption-based separation processes have been extensively applied in the chemical industry for critical separations, as well as emerging separations in the fine chemical industry (Tien, 2018). With the growing emphasis on green technology, it has also emerged as a key solution for environmental control (Noll, 1991).

At the same time, the vast array of potential adsorbents has made adsorption an ever-evolving technology that continues to attract significant attention from both academic and industrial sectors. Zeolites, in particular, stand at the forefront of this technological advancement. These microporous crystalline aluminosilicates possess well-defined pore sizes, enabling them to selectively discriminate between molecules of significant practical interest. Furthermore, the structural and compositional diversity of zeolites allows for the fine-tuning of their adsorption properties, making them exceptionally well-suited for addressing specific separation challenges in an environmentally conscious manner (Pérez-Botella et al., 2022).

Zeolites' framework density (FD) is a critical factor in determining their effectiveness in adsorption-based separation processes. Raman (2023) defines the FD of zeolites as the number of tetrahedrally coordinated atoms (T-atoms) per 1000 $Å^3$ and it is an important index for evaluating the structure-void ratio of a zeolite. In comparison with those of other microporous materials, a zeolite framework is built exclusively from TO_4 tetrahedra (T denotes tetrahedrally coordinated Si, Al, or P). Each TO_4 tetrahedron is connected with four neighbors by sharing their vertex O atoms, forming the three-dimensional zeolite framework. Although all zeolites are constructed from TO_4 tetrahedra, the different ways in which they can be connected lead to the rich variety of zeolite structures (Li, 2014). The structural description of zeolites is typically performed in terms of their building units, with the TO_4 tetrahedra as the primary components. These can be linked following different arrangements, which result in secondary building units (SBUs), CBUs, or "tiles". Originally, SBUs, containing up to 16 T atoms, were used to describe zeolite structures, with 23 types listed in the Database of Zeolite Structures (Baerlocher et al., n.d.). However, since 2007, it has been recognized that SBUs alone are insufficient, leading to the adoption of the broader concepts of CBUs and tiles for a more comprehensive description of zeolite frameworks (Pérez-Botella et al., 2022). Thanks to the development of synthetic strategies, the Structure Commission of the IZA has approved 251 distinct zeolite framework types to date, each of which was assigned a three-letter code. (Baerlocher et al., n.d.) Several zeolite studies have found success using data science to predict the zeolite framework type from crystallographic data (Carr et al., 2009; Yang et al., 2009) and modeling the mechanical properties of zeolites (Evans & Coudert, 2017). However, global methodologies for predicting structural properties of zeolites are still limited.

In the molecular field, advanced techniques in cheminformatics are used to predict molecular properties, particularly through the use of molecular fingerprints such as the Morgan fingerprints (Ding et al., 2021; Fang et al., 2023; Zhong & Guan, 2023). These fingerprints capture essential structural features of molecules for property prediction and serve as input for SciML models. A notable example of this approach is found in the work by Coley et al. (2017), who developed a convolutional neural network (CNN) model for predicting physical properties of molecules. However, despite the success in molecular systems, there have not yet been studies that attempt to transpose such fingerprinting techniques to zeolites. This gap represents a potential area of development, as applying similar methodologies could provide new insights into predicting the structural properties of zeolites, and maybe other properties in the future.

To further enhance the development and synthesis of new zeolites, predicting structural properties—such as FD—using CBUs and SciML offers significant advantages. A particularly promising approach involves creating fingerprints based on the structural characteristics of CBUs, akin to the Morgan fingerprints used in cheminformatics. These CBU-based fingerprints capture the essential structural features of zeolites, serving as input data for ML models that predict key properties like FD. This approach enables the rapid screening of known zeolite structures before synthesis, significantly reducing the time and resources required in experimental trials. On the other hand, it allows for the optimization of zeolite properties for specific adsorption processes, such as gas separation or catalysis, by tailoring the FD to maximize performance. As this methodology is refined, it could lead to faster, more reliable identification of the best candidates for specific adsorption processes, pushing forward advancements in zeolite research and industrial application.

3 Methodology

The proposed methodology consists of three different techniques to achieve the goal of predicting the FD of zeolites based on their CBUs. These were: the creation of the zeolites' database, the creation of the CBU fingerprints and the creation and implementation of an Artificial Neural Network (ANN) to predict the FD values based on the CBU fingerprints created. The scheme presented in Figure 3-1 illustrates the applied methodology detailed in this section.

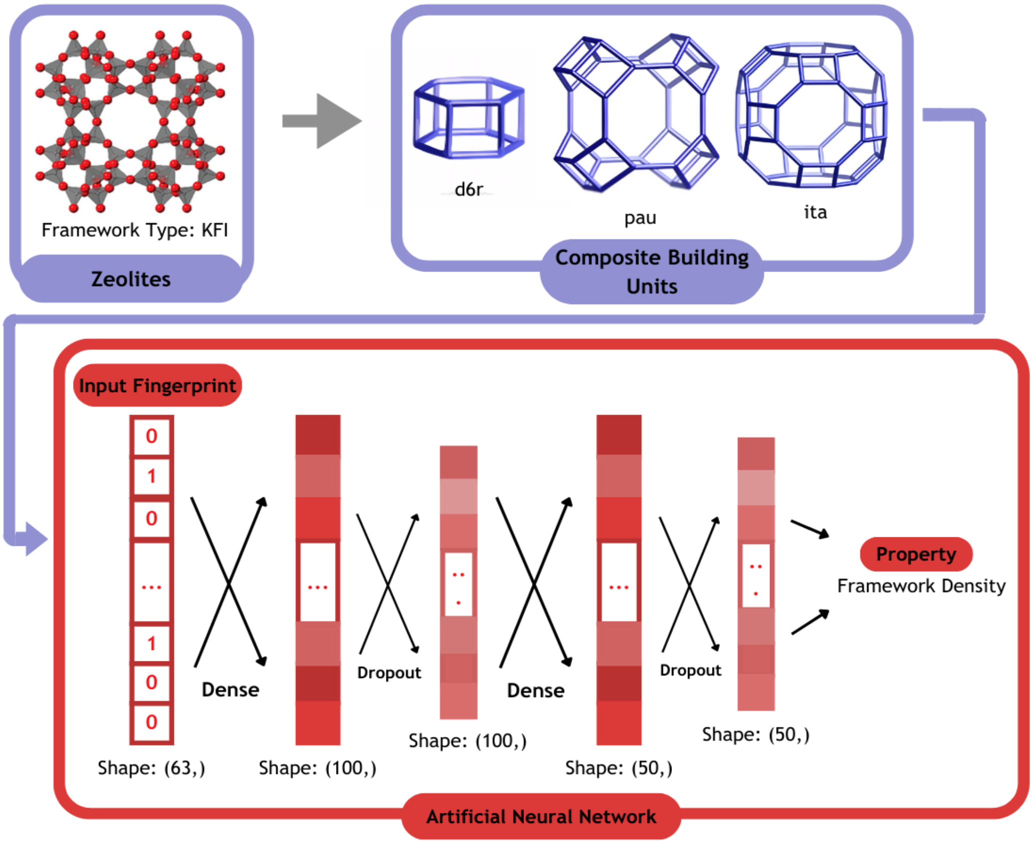

Figure 3-1 - Methodology's scheme

The following subsections will explain each part of the proposed methodology.

3.1 Database

The database used in developing this work was extracted from the IZA website. (Baerlocher et al., n.d.) It was essential to develop a research methodology to look for the required information within the website's HTML code and the programming of a Python language code for that purpose. The last one was developed on the Google Collaboratory platform. A database was obtained for 251 zeolites, containing the following information: Zeolite Type, Unit Cell Shape, Space Group, the lengths a, b and c, the amplitudes α, β and γ, FD, Ring Sizes, Channels dimensions, Maximum diameter of a sphere that can be included, Maximum diameter of a sphere that can diffuse along, Accessible Volume, Natural Tiling, CBUs, Chemical Formula and a column for additional information. A singular database containing the list of T-atoms for each zeolite was also created.

With the database ready, the next crucial step was to define the inputs for the predictive model. Since the goal is to predict zeolites' structural properties, the key input selected was the CBU-based fingerprints for each zeolite. These fingerprints are critical to the model's success, as they capture the unique structural characteristics of each zeolite in a format that is digestible by machine learning algorithms. Generated through a custom-developed code, each fingerprint is represented as a sparse bit vector, with the indices corresponding to the presence or absence of specific CBUs within the zeolite framework.

The fingerprints play a pivotal role in translating complex structural data into a form that can be used for accurate predictions. Each fingerprint consists of 63 bits, one for each of the 63 CBU types listed in the Database of Zeolite Structures (Baerlocher et al., n.d.). This concise yet comprehensive representation of the zeolite structure forms the foundation of the predictive model, allowing it to efficiently process and analyze vast amounts of structural data. Figure 3-2 provides an example of a generated fingerprint, clearly illustrating how the CBUs of a specific zeolite are encoded into this critical input format.

Zeolite Types	lov	d3r	nat	vsv	mei	d4r	mor	sti	bea	bre	jbw	mtt
ABW	0	0	0	0	0	0	0	0	0	0	0	0
ACO	0	0	0	0	0	1	0	0	0	0	0	0
AEI	0	0	0	0	0	0	0	0	0	0	0	0
AEL	0	0	0	0	0	0	0	0	0	0	0	0

Figure 3-2 - Illustrative example of the fingerprints generated to the first zeolites, in alphabetical order

3.2 Artificial Neural Network

ANNs are computational models inspired by the human brain, designed to recognize patterns and learn from data. They consist of interconnected nodes (neurons) organized into layers: an input layer, hidden layers, and an output layer. The connections between nodes have associated weights, which are adjusted during training to minimize prediction errors.

The model is implemented in Python using the ML library Keras (Team, n.d.) built on Theano. (*Welcome — Theano 0.8.2 Documentation*, n.d.) The development of this property prediction method is based on the conclusions of the work of Coley et al. (2017).

3.2.1 Model Architecture

The structure of the model architecture is defined through hyperparameters. Those variables are set as a means to guide and direct the training and performance of the SciML model. The definition of these hyperparameters has major implications on the accuracy of the methodology applied. In this work, they were defined through a trial-and-error method. The defined hyperparameters are presented in Table 1.

The input to the model is the resulting fingerprint represented as a vector of size 63. The input layer does not perform any computation; it simply feeds the input data to the subsequent layers.

The model created contains two hidden layers. These layers perform the bulk of the computation, learning complex patterns in the input data through weighted connections. The first hidden layer has 100 neurons, which apply the tanh activation function. The activation function introduces non-linearity into the model, enabling it to capture complex relationships between the features.

After the first hidden layer, a dropout layer is applied to prevent overfitting. The dropout layer randomly sets a fraction of the neurons (specified by the dropout rate, which in this case is 0.25), to zero during training, forcing the model to learn more robust features.

The second hidden layer has 50 neurons and also applies the same activation function as the first hidden layer. It further refines the learned features. Another dropout layer follows this layer, with a dropout rate of 0.25.

The output layer is a dense layer with size 1, representing the prediction output of the model. The activation function for this layer is linear to produce continuous values.

The model is compiled with the Mean Squared Error (MSE) loss function, which measures the discrepancy between the predicted and actual values. The optimizer implemented is the Adam optimizer, which uses a learning rate of 0.001 to minimize the loss by adjusting the weights through backpropagation.

Table 1 - Artificial Neural Network model hyperparameters.

Parameters	Neural Network Models's Values
Learning Rate	1.00×10^{-3}
Epochs	25 per fold
Activation Function (Hidden Layers)	Tanh
Dropout Probability (Hidden Layers)	0.25
Hidden Layer 1 Dimension	100
Hidden Layer 2 Dimension	50
Loss Function	Mean Squared Error (MSE)
Output Layer Activation Function	Linear
Total Number of Layers	3 layers (2 hidden + 1 output)
Number of Samples	251 (full dataset)
Optimizer	Adam
Early Stopping Patience	10 epochs
Validation Split	10% from training set
Cross-Validation Folds	5

3.2.2 Training Process

The data is divided into three sets: training, validation, and testing. The training set is used to adjust the model weights, while the validation set monitors performance during training to prevent overfitting. The test set evaluates the final model performance after training. In this method, the 5-fold cross-validation (CV) was used to divide, train and test my data. In this 5-fold CV, the data is split into 5 folds. For each CV fold, one fold is used as the test set, and the remaining four folds are combined to form the training set. From the samples in the training set, a random 10% is set aside as the validation set. (Galeazzo & Shiraiwa, 2022)

The model also incorporates an early stopping mechanism with patience specified to 10. This stops training if the validation loss does not improve for a specified number of epochs (in this case, 10 epochs), preventing overfitting and saving computational resources.

The model is trained for 25 epochs, for each fold. The model is trained using the fit method, which iteratively adjusts the weights based on the training data. The training process can be interrupted manually if necessary.

3.2.3 Model Evaluation and Testing

After training, the model is evaluated on the test set to assess its performance. The absolute and relative error for each testing sample are also determined. Additionally, a parity plot is generated to visualize the relationship between predicted and actual values, providing insight into the model's accuracy. The parity plot compares the true vs. predicted values. The closer the points are to the diagonal line, the better the model's performance. This plot helps identify any systematic errors or outliers.

The architecture, weights, and training history of the resulting model are then saved to disk for future use. The test performance is recorded, and the model's predictions are analyzed for accuracy.

4 Results and Discussion

The model was trained for 25 epochs, for each fold. The training was performed in a Windows 11 environment, through the Windows Subsystem for Linux. The system has an AMD Ryzen 5 5500U with Radeon Graphics 2.10 GHz; 8.0 GB of installed RAM; an operative system of 64 bits and an AMD Radeon(TM) Graphics GPU.

Figure 4-1 presents part of the zeolite database that was created, in Excel format, with some of the properties of interest. From Figure 4-2, it is possible to visualize one of the tables referring to the list of T-atoms of a zeolite, also in Excel format.

	Zeolite Types	Unit Cell Shape	Space Group	a (Å)	b (Å)	c (Å)	α (°)	β (°)	γ (°)
0	ABW	orthorhombic	I m m a	9.8730	5.2540	9.8730	90.000°	90.000°	90.000°
1	ACO	cubic	I m -3 m	9.9050	9.9050	9.9050	90.000°	90.000°	90.000°
2	AEI	orthorhombic	C m c m	13.6770	12.6070	13.6770	90.000°	90.000°	90.000°
3	AEL	orthorhombic	I m m a	8.3120	18.7290	8.3120	90.000°	90.000°	90.000°
4	AEN	orthorhombic	C m c e	18.5310	13.3550	18.5310	90.000°	90.000°	90.000°
5	AET	orthorhombic	C m c m	32.8290	14.3800	32.8290	90.000°	90.000°	90.000°

Figure 4-1 - Illustrative example of the database related to the zeolites' properties of interest, obtained through web scrapping of the IZA website

	A	B	C	D	E	F	G
1	T-atom Name	Site Multiplicity	x	y	z	Symmetry Restrictions	Site Symmetry
2	T1	16	0.204	0.3567	0.0614	X,Y,Z	1
3	T2	8	0.5	0.0599	0.0572	1/2,Y,Z	m
4	T3	8	0.5	0.225	0.0681	1/2,Y,Z	m
5	T4	8	0	0.1636	0.0647	0,Y,Z	m
6							

Figure 4-2 - Illustrative example of the database related to the list of T-atoms of zeolite AFO, obtained through web scrapping of the IZA website

With part of the content of this database, the CBU-based fingerprint for each zeolite was generated. Figure 4-3 presents part of the CBU fingerprints that were created, in Excel format.

Zeolite Types	lov	d3r	nat	vsv	mei	d4r	mor	sti	bea	bre	jbw	mtt
ABW	0	0	0	0	0	0	0	0	0	0	0	0
ACO	0	0	0	0	0	1	0	0	0	0	0	0
AEI	0	0	0	0	0	0	0	0	0	0	0	0
AEL	0	0	0	0	0	0	0	0	0	0	0	0
AEN	0	0	0	0	0	0	0	0	0	0	0	0
AET	0	0	0	0	0	0	0	0	0	0	0	0
AFG	0	0	0	0	0	0	0	0	0	0	0	0
AFI	0	0	0	0	0	0	0	0	0	0	0	0
AFN	0	0	0	0	1	0	0	0	0	0	0	0
AFO	0	0	0	0	0	0	0	0	0	0	0	0

Figure 4-3 - Illustrative example of the input fingerprints related to the zeolites' CBUs, obtained through Python techniques

As such, the resulting fingerprint representation is passed through additional hidden layers in a feed forward neural network to yield a single scalar value (which is FD, in this case). In Figure 4-4, it is possible to see the ANN model results, corresponding to a fold in CV. The parity plots for all the 5 folds of the 5-fold CV method used may be consulted separately in Appendix B. In Appendix A, it is possible to find some details regarding the predictions of each individual zeolite of the test set for all the 5 folds in one training, including relative and absolute errors between each actual and predicted value.

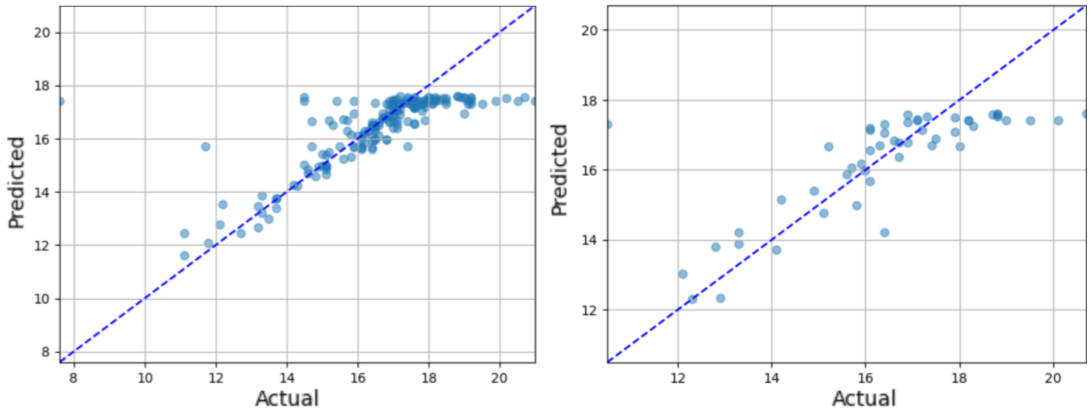

Figure 4-4 - Parity plots between predicted and actual values of the Zeolites' Framework Density (T/1000 Å^3) for: a) the training set; b) the test set

Regarding the results, it was possible to visualize the presence of outliers in the model's predictions, with a relative error superior to 10% to the actual value. It is important to consider possible reasons for these discrepancies. One potential reason is the low representation of certain zeolite framework densities within the training data. In Figure 4-5, The frequency distribution of zeolite framework densities reveals a limited representation of certain ranges, particularly for zeolites with low framework densities and high framework densities. This imbalance in the data may contribute to the model's reduced performance for these specific frameworks, as it has encountered fewer examples within these ranges during training, hindering its ability to accurately predict their properties.

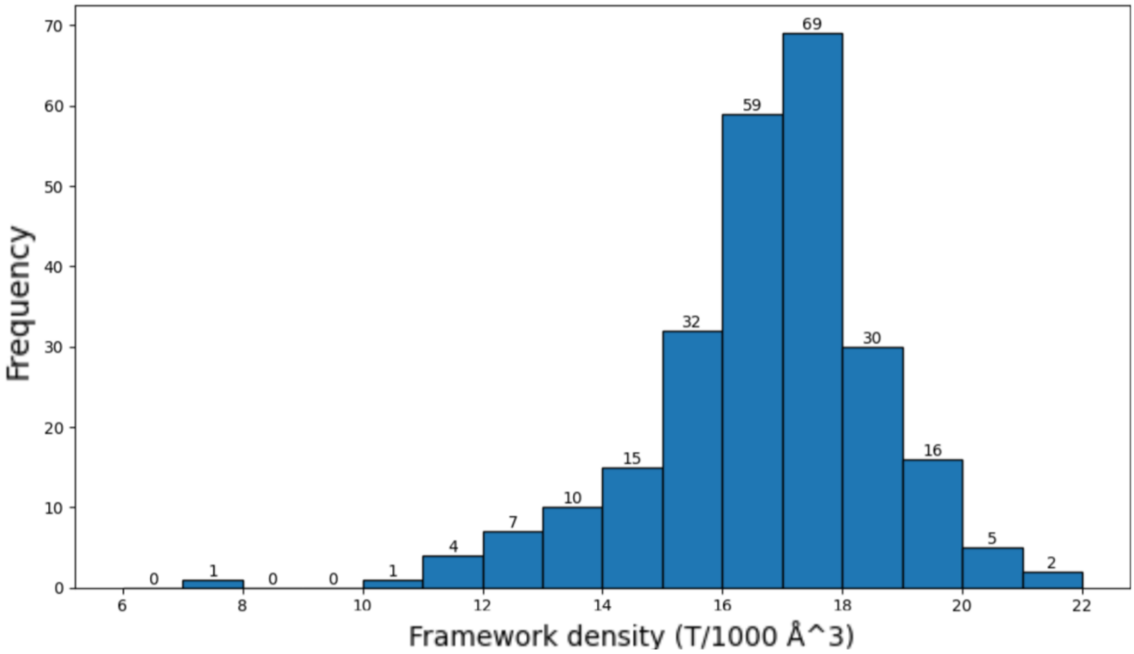

Figure 4-5 - Frequency Distribution of Zeolite Framework Densities

Additionally, another factor contributing to the outliers could be the structural complexity of some zeolites. Some of the outliers did not have any CBU expression for their structure. Instead, their structure was only accurately described using natural tiling, another form of structural expression for zeolites. According to Čejka et al. (2017), natural tiling captures the arrangement of tiles in the zeolite framework and offers an unambiguous and consistent method for describing zeolite frameworks. Unlike CBUs, which can be assigned in multiple ways, leading to potential overlap and ambiguity, natural tiling provides a complete and definitive breakdown of the framework. Tiles, also known as natural building units (NBUs), divide the Euclidean space of the framework and always share faces with exactly two other tiles, ensuring that the entire space is accounted for without gaps. This comprehensive coverage contrasts with CBUs, which may leave parts of the structure unrepresented.

Moreover, natural tiling is applicable to any zeolite framework, regardless of its complexity, ensuring that even the most intricate structures can be consistently described. Interestingly, while no zeolites have a CBU description without a corresponding natural tiling description, the reverse is true—some zeolites only have a natural tiling expression and lack a CBU description. This suggests that natural tiling may be a more universally applicable method for representing zeolite structures.

Given these advantages, it may be worthwhile to explore in future work whether using natural tiling fingerprints instead of CBU-based fingerprints leads to better prediction accuracy. By incorporating natural tiling fingerprints, it may be possible to enhance the model's ability to predict FD, especially for those zeolites where CBUs are not well-represented.

Despite the presence of some outliers in the predictions, it is crucial to highlight the model's overall success, particularly in accurately predicting the framework densities of zeolites with well-represented CBUs. The artificial neural network (ANN) consistently demonstrated high accuracy for zeolites where CBU-based fingerprinting could effectively capture the structural nuances. For these zeolites, the relative errors were often below 10%, reflecting the strength of the proposed methodology when applied to well-represented structures. This showcases the reliability of the SciML approach, which leverages CBU fingerprints as a powerful tool for predicting zeolite properties in a much faster and resource-efficient manner than traditional experimental techniques.

5 Conclusions

This project tackled the challenge of predicting the framework density (FD) of zeolites, a crucial parameter for optimizing their performance in chemical separation processes. Innovation was key at every stage, starting with the novel application of Scientific Machine Learning (SciML) techniques to zeolite structures. Significant advancements were made through the creation of a large-scale database via web scraping and the development of composite building unit (CBU)-based fingerprints, which systematically characterized zeolite structures for prediction purposes.

The results demonstrated the strong potential of this approach, with the model accurately predicting the FD of numerous zeolites. The comprehensive database, generated from the IZA website, was essential in producing the unique CBU fingerprints that served as the model’s foundation. These fingerprints enabled precise predictions of structural characteristics, showcasing the robustness of the model not only in predicting FD but also in offering valuable insights into the broader properties of the materials.

However, the presence of outliers in the model's predictions highlighted certain limitations. These discrepancies were primarily attributed to the low representation of certain framework densities in the training data and the structural complexity of some zeolites, which were not fully captured by CBU-based fingerprints. In particular, some zeolites lacked a CBU expression and were more accurately described using natural tiling, a method that offers an unambiguous and complete representation of the zeolite framework. Given these findings, future work should explore the use of natural tiling fingerprints as an alternative or complementary approach to improve the model's accuracy.

The solution developed is highly beneficial to the institution hosting the internship. The database facilitates faster data extraction and comparison of existing zeolites, while the model reduces the time and resources needed for experimental trials. This enables rapid screening and optimization of zeolites for industrial applications, advancing research and development efforts toward more sustainable and efficient processes. By integrating advanced ML with chemical engineering, the institution positions itself at the forefront of innovation, contributing to its strategic goals.

6 Evaluation of the performed work

6.1 Objectives Achieved

This work introduces an innovative approach to predicting the framework density (FD) of zeolites by combining Scientific Machine Learning (SciML) with chemical engineering. A predictive model using artificial neural networks (ANNs) was successfully developed, leveraging CBU-based fingerprints from a comprehensive zeolite database. This approach offers a more efficient, resource-saving alternative to traditional methods, showing strong predictive accuracy for most zeolites while identifying areas for improvement. Overall, the project advanced the use of SciML in chemical engineering and provided valuable insights for enhancing prediction accuracy.

6.2 Final Assessment

The development of this work was a deeply enriching experience on a personal level, as it allowed me to contribute to the intersection of machine learning and chemical engineering in a meaningful way. The project was undoubtedly challenging, requiring the integration of advanced SciML techniques with the complex domain of zeolite chemistry. Through this process, I gained invaluable insights into the power of programming and data science in solving real-world problems, particularly in the context of predicting zeolite FD.

6.3 Framing with the ODS

The contribution of this work to the Sustainable Development Goals (SDGs) can be considered positive, particularly in relation to SDG 9 (Industry, Innovation, and Infrastructure) and SDG 12 (Responsible Consumption and Production). By developing a predictive model that leverages Scientific Machine Learning (SciML) to optimize the framework density (FD) of zeolites, this project directly supports the advancement of sustainable industrial processes. The ability to accurately predict zeolite properties reduces the need for resource-intensive experimental methods, thereby promoting more efficient use of materials and energy in the chemical industry.

Moreover, the work contributes to SDG 13 (Climate Action) by facilitating the development of environmentally friendly separation technologies. Zeolites are integral to many industrial processes, including those aimed at reducing greenhouse gas emissions and improving energy efficiency.

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Appendix A - Zeolite Framework Density Prediction Accuracy and Outlier Analysis

A.1 Test Set of First Fold in 5-Fold Cross-Validation

Table 2 - Model Performance on Test Set of first fold in Cross-Validation

Test Entry	Zeolite Type	Actual Value	Predicted Value	Actual - Predicted	Relative Error(%)
0	EWO	19.1	18.22478	0.87522	4.58230
1	UOS	17.6	17.44457	0.15543	0.88313
2	AFO	19.2	18.23851	0.96149	5.00777
3	VFI	14.5	18.23851	-3.73851	25.78281 >10 --- outlier
4	OWE	17.1	16.86591	0.23409	1.36893
5	CHA	15.1	15.03200	0.06800	0.45030
6	MFS	17.4	17.77073	-0.37073	2.13062
7	-EWT	14.5	18.08628	-3.58628	24.73294 >10 --- outlier
8	IFY	17.6	17.38971	0.21029	1.19481
9	STW	16.4	16.62506	-0.22507	1.37235
10	ESV	17.7	18.08628	-0.38628	2.18235
11	MFI	18.4	18.23882	0.16118	0.87598
12	DOH	17.0	17.94800	-0.94800	5.57648
13	THO	15.7	16.00693	-0.30693	1.95497
14	USI	15.9	18.16395	-2.26395	14.23869 >10 --- outlier
15	MAZ	16.7	17.73167	-1.03167	6.17764
16	NPT	13.5	13.99880	-0.49880	3.69484
17	EON	16.8	17.59500	-0.79500	4.73213
18	PWW	17.0	17.76680	-0.76680	4.51057
19	-CHI	19.9	18.08628	1.81372	9.11418
20	IMF	17.5	17.45259	0.04741	0.27091
21	SBS	13.7	13.95484	-0.25484	1.86013
22	DON	17.1	18.16326	-1.06326	6.21788
23	BCT	19.0	18.16164	0.83836	4.41240
24	LEV	15.9	16.75259	-0.85259	5.36219
25	AVL	15.8	14.82304	0.97696	6.18331
26	SAT	16.4	16.80309	-0.40309	2.45786
27	SFO	15.2	16.86591	-1.66591	10.95996 >10 --- outlier
28	EAB	16.0	15.31252	0.68748	4.29673
29	HEU	17.5	17.76680	-0.26680	1.52455
30	MER	16.4	16.84768	-0.44768	2.72973
31	CZP	21.2	18.08628	3.11372	14.68738 >10 --- outlier
32	UFI	15.2	15.05886	0.14114	0.92854
33	SFH	16.5	17.77345	-1.27345	7.71785
34	TOL	17.9	18.03358	-0.13358	0.74627

35	MEI	14.7	17.82864	-3.12864	21.28327 >10 --- outlier
36	ATT	17.1	17.85258	-0.75258	4.40106
37	IFO	17.3	18.16164	-0.86164	4.98061
38	EDI	16.3	16.00693	0.29307	1.79797
39	JNT	20.5	18.08628	2.41372	11.77426 >10 --- outlier
40	SSY	16.9	17.37002	-0.47002	2.78120
41	FAR	17.1	17.96054	-0.86054	5.03242
42	GIS	16.4	17.52805	-1.12805	6.87835
43	OFF	16.1	16.24544	-0.14544	0.90336
44	RTE	17.2	17.79584	-0.59583	3.46416
45	FER	17.6	18.01047	-0.41047	2.33222
46	STF	16.9	18.11047	-1.21047	7.16255
47	ITW	17.7	16.62506	1.07494	6.07308
48	SBT	13.7	13.95484	-0.25484	1.86013
49	GOO	19.0	18.08628	0.91372	4.80907
50	AFN	17.4	17.82864	-0.42864	2.46345

A.2 Test Set of Second Fold in 5-Fold Cross-Validation

Table 3 - Model Performance on Test Set of second fold in Cross-Validation

Test Entry	Zeolite Type	Actual Value	Predicted Value	Actual - Predicted	Relative Error(%)
0	STI	16.7	16.55153	0.14847	0.88905
1	PUN	15.0	16.37504	-1.37504	9.16696
2	-ITV	10.5	17.42767	-6.92767	65.97784 >10 --- outlier
3	-IFU	12.1	11.94577	0.15423	1.27464
4	BOF	18.3	17.24242	1.05758	5.77912
5	IFR	17.2	17.69232	-0.49232	2.86231
6	AHT	19.2	17.88437	1.31563	6.85225
7	IWW	16.6	15.89785	0.70215	4.22981
8	PTT	16.5	16.28835	0.21165	1.28271
9	SAV	14.6	14.70208	-0.10208	0.69919
10	SZR	17.6	17.60674	-0.00674	0.03830
11	-HOS	14.7	14.47475	0.22525	1.53229
12	IWS	14.8	14.50366	0.29634	2.00228
13	SFN	16.6	17.36095	-0.76095	4.58401
14	JOZ	16.4	14.57870	1.82130	11.10547 >10 --- outlier
15	MTT	18.2	17.82870	0.37130	2.04009
16	TER	17.0	17.22118	-0.22118	1.30104
17	SFS	16.6	16.52319	0.07681	0.46274
18	AFI	16.9	17.95250	-1.05250	6.22783
19	MTN	17.2	17.31498	-0.11497	0.66846
20	-WEN	16.6	16.62348	-0.02348	0.14143
21	MAR	17.6	17.45941	0.14059	0.79882
22	UEI	17.5	17.73372	-0.23372	1.33556
23	CAS	18.8	17.95788	0.84212	4.47937

24	SWY	16.1	15.63575	0.46425	2.88357
25	SOR	17.1	17.14536	-0.04536	0.26526
26	POR	17.6	17.10346	0.49654	2.82124
27	ATN	17.8	17.60846	0.19154	1.07609
28	JST	14.3	14.57870	-0.27870	1.94897
29	AVE	16.5	15.83702	0.66298	4.01808
30	PCR	18.8	17.96628	0.83372	4.43469
31	NPO	15.9	17.73372	-1.83372	11.53285 >10 --- outlier
32	SEW	17.0	17.09611	-0.09611	0.56536
33	-RON	19.1	17.82984	1.27016	6.65005
34	YFI	16.4	16.50372	-0.10372	0.63245
35	MSO	17.8	17.44985	0.35015	1.96714
36	ATO	18.8	17.86958	0.93042	4.94904
37	CON	15.7	16.22916	-0.52916	3.37047
38	IWV	15.0	15.74900	-0.74900	4.99331
39	NAB	16.1	16.37504	-0.27504	1.70834
40	ITH	17.4	14.81435	2.58565	14.86008 >10 --- outlier
41	CGS	16.9	17.73372	-0.83372	4.93327
42	SIV	16.4	15.47858	0.92142	5.61844
43	WEI	16.5	16.37504	0.12496	0.75731
44	-IRY	11.1	10.92456	0.17544	1.58055
45	MEP	17.9	17.31498	0.58502	3.26829
46	LOS	16.9	17.36859	-0.46859	2.77271
47	OSI	17.7	17.48636	0.21364	1.20700
48	AFX	15.1	15.05504	0.04496	0.29774
49	EZT	17.2	17.11162	0.08838	0.51386
50	ITT	12.8	13.07309	-0.27309	2.13355

A.3 Test Set of Third Fold in 5-Fold Cross-Validation

Table 4 - Model Performance on Test Set of third fold in Cross-Validation

Test Entry	Zeolite Type	Actual Value	Predicted Value	Actual - Predicted	Relative Error(%)
0	BOZ	12.9	13.60428	-0.70428	5.45953
1	VSV	16.8	17.09467	-0.29467	1.75399
2	AWO	18.2	17.93537	0.26463	1.45400
3	ANO	16.5	16.07900	0.42100	2.55150
4	UOZ	19.5	15.70922	3.79078	19.43988 >10 --- outlier
5	FAU	13.3	13.93006	-0.63006	4.73727
6	MEL	17.4	17.86799	-0.46799	2.68960
7	ITE	15.7	15.85486	-0.15486	0.98639
8	CDO	18.1	17.71137	0.38863	2.14712
9	APD	18.0	17.93537	0.06463	0.35904
10	AFR	15.2	16.62545	-1.42545	9.37796
11	AET	18.2	18.07356	0.12644	0.69474

12	JBW	18.8	18.10572	0.69428	3.69297
13	PTO	18.2	17.93537	0.26463	1.45400
14	CAN	16.9	17.84109	-0.94109	5.56856
15	TSC	13.2	13.75280	-0.55280	4.18785
16	ACO	16.5	15.75480	0.74520	4.51638
17	MTF	20.7	18.10847	2.59153	12.51945 >10 --- outlier
18	ETV	17.3	17.07917	0.22083	1.27647
19	MRT	17.1	17.34690	-0.24690	1.44387
20	RTH	16.1	15.85486	0.24514	1.52259
21	-SYT	12.2	12.46871	-0.26871	2.20253
22	EEI	17.9	17.99107	-0.09107	0.50876
23	SBN	16.1	17.80654	-1.70654	10.59961 >10 --- outlier
24	OSO	13.3	16.03966	-2.73966	20.59897 >10 --- outlier
25	-IRT	11.7	14.86188	-3.16188	27.02460 >10 --- outlier
26	MOZ	17.0	17.07797	-0.07797	0.45863
27	ZON	18.0	16.62545	1.37455	7.63639
28	UWY	16.3	16.24412	0.05588	0.34280
29	AFT	15.1	15.79925	-0.69925	4.63079
30	GME	15.1	15.85946	-0.75946	5.02955
31	ETR	15.4	17.93537	-2.53537	16.46347 >10 --- outlier
32	RWR	19.2	17.79688	1.40313	7.30795
33	RHO	14.5	14.27898	0.22102	1.52429
34	TON	18.1	18.00361	0.09639	0.53256
35	PWN	15.6	15.31485	0.28516	1.82792
36	BOG	16.1	17.37313	-1.27313	7.90762
37	LTN	17.0	17.21147	-0.21147	1.24395
38	ISV	15.0	14.91937	0.08063	0.53754
39	EPI	17.7	17.79688	-0.09687	0.54731
40	EUO	17.1	17.99107	-0.89107	5.21092
41	LAU	18.0	17.98078	0.01922	0.10677
42	GON	17.7	17.50437	0.19563	1.10528
43	LTA	14.2	14.77189	-0.57189	4.02737
44	MWW	15.9	15.89314	0.00686	0.04313
45	SFW	15.1	15.85946	-0.75946	5.02955
46	JZO	13.2	13.90470	-0.70470	5.33863
47	EWF	17.9	17.64734	0.25266	1.41149
48	JSR	12.3	13.29018	-0.99018	8.05026
49	PON	18.2	17.93537	0.26463	1.45400
50	MTW	18.2	17.87320	0.32680	1.79559

A.4 Test Set of Forth Fold in 5-Fold Cross-Validation

Table 5 - Model Performance on Test Set of fourth fold in Cross-Validation

Test Entry	Zeolite Type	Actual Value	Predicted Value	Actual - Predicted	Relative Error(%)
0	PAU	15.9	16.32854	-0.42854	2.69523
1	KFI	15.0	14.28503	0.71497	4.76646
2	RRO	17.9	17.41338	0.48662	2.71855
3	AEI	15.1	15.28288	-0.18288	1.21111
4	RUT	18.1	17.90388	0.19612	1.08351
5	NES	16.4	17.03483	-0.63483	3.87094
6	AFG	16.9	17.44388	-0.54388	3.21825
7	SAS	14.9	15.28288	-0.38288	2.56965
8	ETL	18.0	17.90222	0.09778	0.54323
9	NAT	16.2	15.63380	0.56620	3.49505
10	PWO	17.1	17.41338	-0.31338	1.83263
11	STT	17.0	17.07055	-0.07055	0.41503
12	SOD	16.7	17.10835	-0.40834	2.44518
13	ITG	16.6	16.28708	0.31292	1.88505
14	IRN	15.3	16.31878	-1.01878	6.65871
15	OKO	17.5	16.10017	1.39983	7.99901
16	AEN	20.1	17.72145	2.37855	11.83358 >10 --- outlier
17	NON	17.6	17.74117	-0.14117	0.80213
18	ANA	19.2	17.72145	1.47855	7.70078
19	ABW	17.6	17.88634	-0.28634	1.62696
20	IRR	11.8	10.45191	1.34809	11.42446 >10 --- outlier
21	OBW	12.7	12.35017	0.34983	2.75458
22	LTF	16.9	16.95705	-0.05705	0.33756
23	ATV	18.9	17.88775	1.01225	5.35582
24	JSN	17.9	17.32456	0.57544	3.21476
25	MVY	21.0	17.72145	3.27855	15.61214 >10 --- outlier
26	RWY	7.6	17.72145	-10.12145	133.17699 >10 --- outlier
27	IFW	15.8	16.03094	-0.23094	1.46166
28	SFG	17.6	17.07617	0.52383	2.97633
29	AFV	15.7	16.47454	-0.77454	4.93338
30	VET	20.2	17.66221	2.53779	12.56331 >10 --- outlier
31	MSE	16.4	16.87055	-0.47055	2.86922
32	ERI	16.1	16.56383	-0.46383	2.88094
33	SOV	13.3	14.39352	-1.09352	8.22196
34	NSI	18.8	17.91488	0.88512	4.70811
35	DFT	17.7	17.72145	-0.02145	0.12119
36	SFE	16.9	17.84140	-0.94140	5.57042
37	ATS	16.1	15.02614	1.07386	6.66993
38	SAO	14.2	14.34727	-0.14727	1.03711
39	IHW	18.5	17.03483	1.46517	7.91982
40	AEL	19.2	17.88775	1.31225	6.83465

41	LIO	17.6	17.63266	-0.03266	0.18554
42	-CLO	11.1	11.08404	0.01596	0.14376
43	AFS	14.6	14.33410	0.26590	1.82125
44	PTY	17.0	17.41338	-0.41338	2.43165
45	SOS	17.0	17.72145	-0.72145	4.24383
46	SSF	16.5	17.16580	-0.66580	4.03516
47	UTL	15.6	14.71634	0.88366	5.66448
48	FRA	17.1	17.41087	-0.31087	1.81797
49	SAF	19.0	17.88775	1.11225	5.85395
50	BPH	14.6	14.96531	-0.36531	2.50211

A.5 Test Set of Fifth Fold in 5-Fold Cross-Validation

Table 6 - Model Performance on Test Set of fifth fold in Cross-Validation

Test Entry	Zeolite Type	Actual Value	Predicted Value	Actual - Predicted	Relative Error(%)
0	JRY	18.5	17.96442	0.53558	2.89505
1	VNI	17.6	17.29012	0.30988	1.76067
2	-PAR	19.5	17.83075	1.66925	8.56026
3	GIU	17.1	17.44597	-0.34597	2.02323
4	UOV	16.2	16.40682	-0.20682	1.27666
5	LTL	16.7	17.25172	-0.55172	3.30371
6	MWF	16.1	15.84722	0.25278	1.57004
7	SFF	17.8	17.55450	0.24550	1.37922
8	PTF	16.0	16.68240	-0.68240	4.26500
9	SGT	17.5	17.83075	-0.33075	1.89000
10	AST	15.8	16.63044	-0.83044	5.25592
11	MOR	17.0	17.72982	-0.72982	4.29306
12	IWR	15.6	15.94457	-0.34457	2.20877
13	MON	17.6	17.83075	-0.23075	1.31107
14	DAC	17.5	17.72982	-0.22982	1.31326
15	-LIT	19.2	17.98247	1.21753	6.34131
16	DFO	14.9	14.56695	0.33305	2.23520
17	CGF	19.0	17.79081	1.20919	6.36414
18	LTJ	18.5	17.83075	0.66925	3.61757
19	YUG	18.0	17.83075	0.16925	0.94028
20	CSV	16.2	16.42246	-0.22246	1.37320
21	-IFT	12.1	13.01498	-0.91498	7.56183
22	PSI	20.7	17.99605	2.70395	13.06255 >10 --- outlier
23	EWS	17.3	17.91196	-0.61197	3.53737
24	SBE	13.7	13.74771	-0.04771	0.34827
25	CFI	16.8	17.57057	-0.77057	4.58675
26	TUN	17.6	17.58957	0.01044	0.05929
27	-SSO	16.7	17.19019	-0.49019	2.93527

28	SOF	16.4	16.95089	-0.55089	3.35911
29	PHI	16.4	17.20339	-0.80339	4.89875
30	RSN	16.8	17.09874	-0.29874	1.77824
31	BSV	18.7	17.83075	0.86925	4.64840
32	-SVR	17.2	17.98104	-0.78104	4.54091
33	BRE	18.3	17.88051	0.41949	2.29230
34	EMT	13.3	13.98511	-0.68511	5.15119
35	AFY	14.1	15.09777	-0.99777	7.07635
36	SVV	18.0	17.97409	0.02591	0.14395
37	ITR	17.4	15.89151	1.50849	8.66948
38	AWW	16.9	17.00630	-0.10630	0.62902
39	DDR	17.9	17.80066	0.09934	0.55496
40	JSW	18.4	17.83075	0.56925	3.09375
41	LOV	16.8	17.09874	-0.29874	1.77824
42	BIK	18.7	17.99544	0.70456	3.76770
43	BEC	15.1	16.13924	-1.03924	6.88241
44	ASV	19.1	16.28610	2.81390	14.73248 >10 --- outlier
45	APC	17.7	17.83075	-0.13075	0.73869
46	POS	15.5	14.01993	1.48007	9.54884

Appendix B - Cross-Validation Parity Plots for Framework Density Prediction

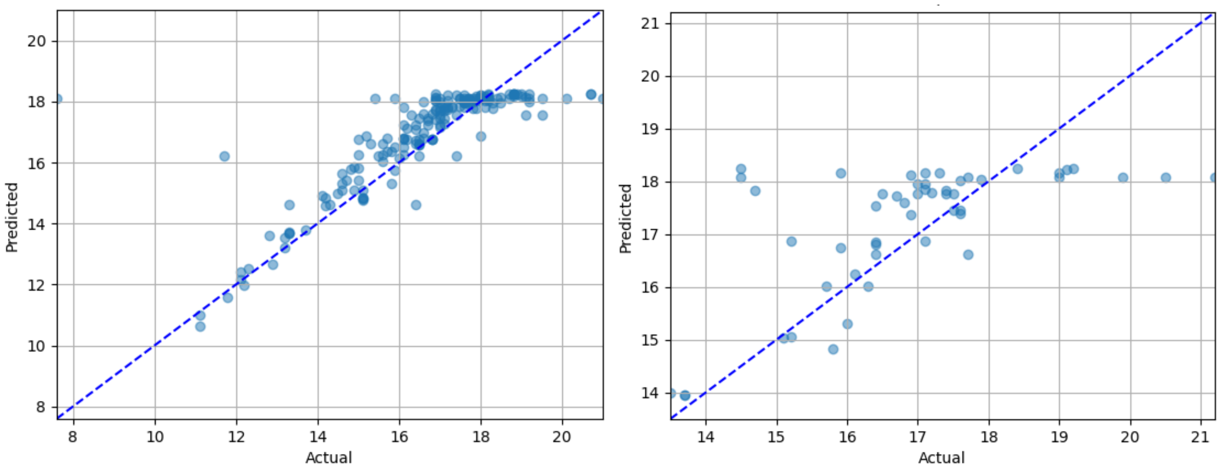

Figure B.1 - Parity plots between predicted and actual values of the Zeolites' Framework Density (T/1000 Å^3) for the first fold in Cross-Validation, for: a) the training set; b) the test set

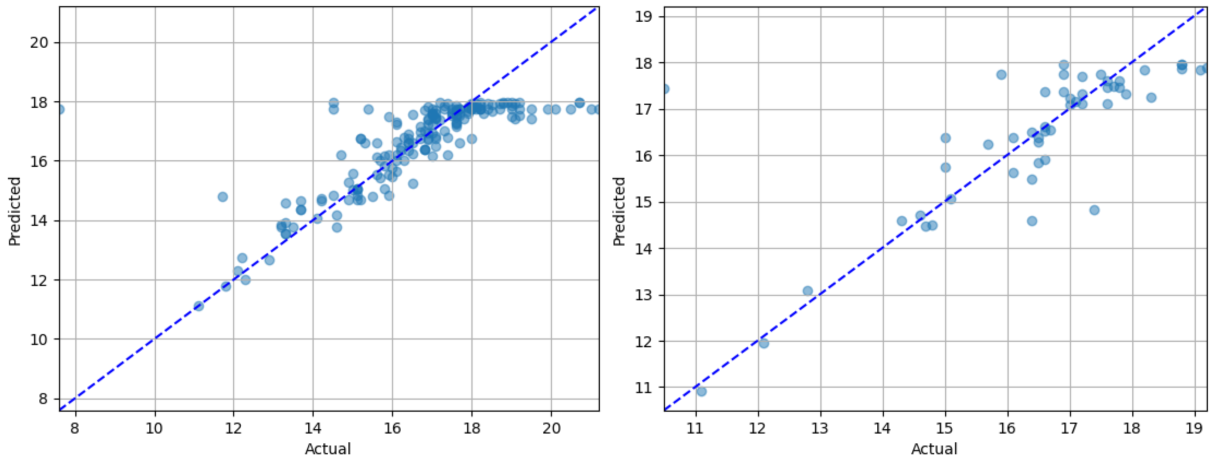

Figure B.2 - Parity plots between predicted and actual values of the Zeolites' Framework Density (T/1000 Å^3) for the second fold in Cross-Validation, for: a) the training set; b) the test set

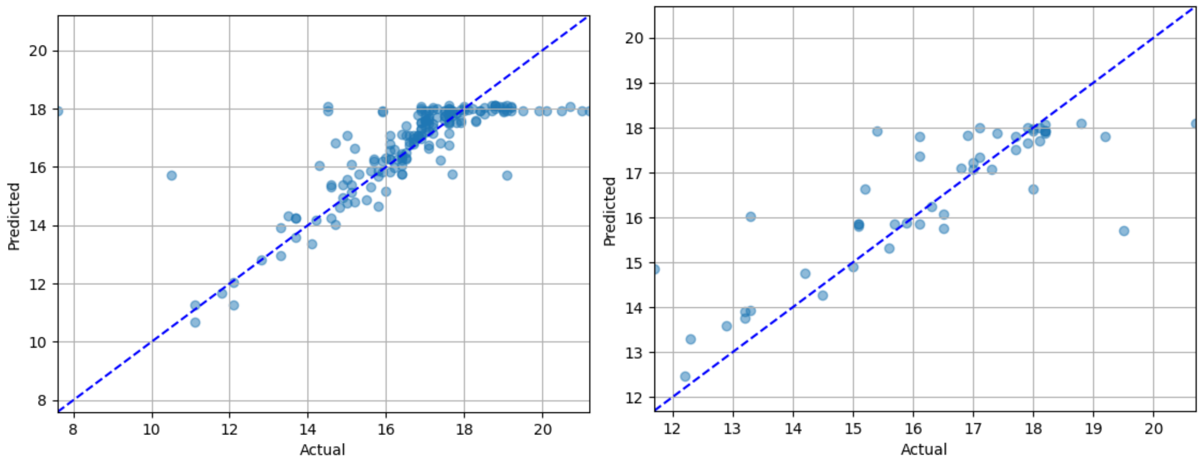

Figure B.3 - Parity plots between predicted and actual values of the Zeolites' Framework Density (T/1000 Å^3) for the third fold in Cross-Validation, for: a) the training set; b) the test set

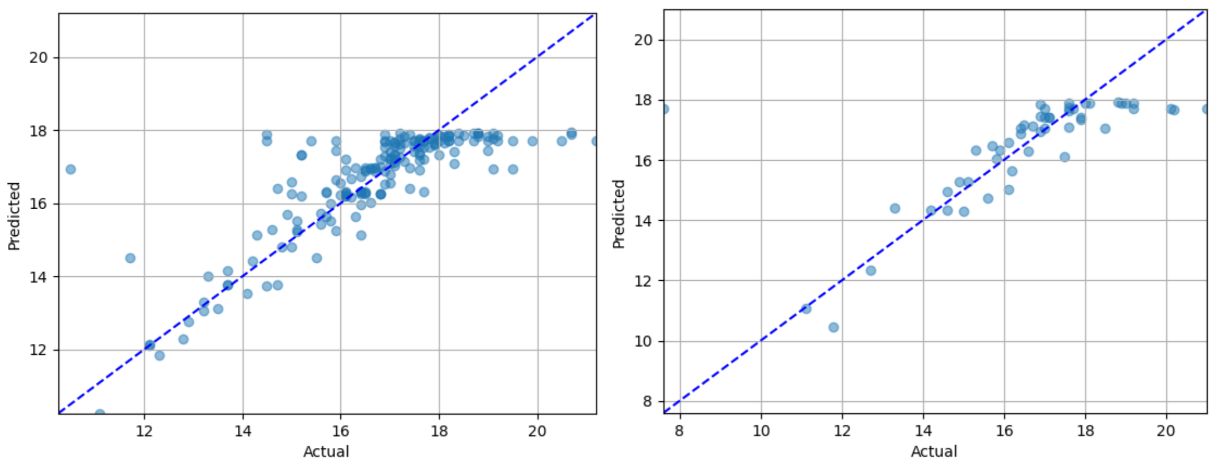

Figure B.4 - Parity plots between predicted and actual values of the Zeolites' Framework Density (T/1000 Å^3) for the fourth fold in Cross-Validation, for: a) the training set; b) the test set

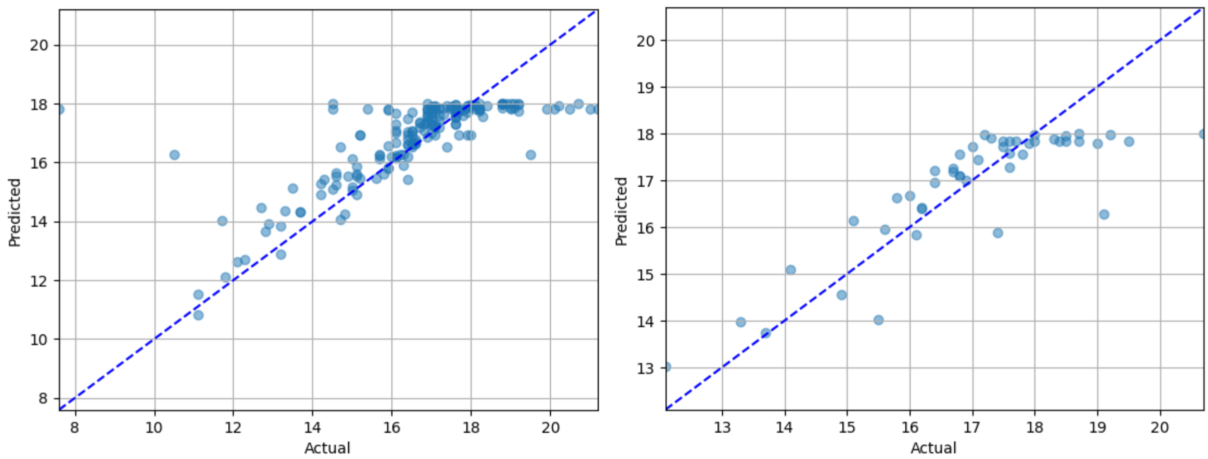

Figure B.5 - Parity plots between predicted and actual values of the Zeolites' Framework Density (T/1000 Å^3) for the fifth fold in Cross-Validation, for: a) the training set; b) the test set